CMSC 104 Section 2
Fall 2026
Classwork #6

Strings and Lists

In-class date: October 21, 2026
Due date: before midnight, October 28, 2026

Assignment:

Part 1: Working With a One-dimensional List

We have a one-dimensional list, `student_record`, which contains in order: a student last name, a student first name, a student ID, and grades for 6 different assignments. Each entry is a string[1]. That is, `student_record` looks like this:

```
['Beard', 'Coach', 'LT1HIA', '54', '41', '70', '58', '13', '12']
['Bumbercatch', 'Moe', '5AEOPC', '59', '42', '65', '69', '14', '173']
['Cartrick', 'George', 'I4K7HQ', '61', '60', '76', '66', '18', '192']
['Cole', 'Cam', 'POSYK0', '74', '58', '66', '69', '87', '63']
['Crimm', 'Trent', 'J7CUK9', '74', '75', '73', '62', '111', '197']
```

Create a new, empty list, called `grade_list`.

Go through each element of `student_record`, and use isdigit() to identify each numeric element that can be converted into an integer. Append that integer value to the end of `grade_list`
Then, sum all of those integer values in `grade_list` to get a total grade for this student. If the student's grade is greater than 540 points, your program should print out something like

[1] Why would scores be strings? Because this has been read in from a file. But that's a lecture from November, so for now, they just are strings.

```
This student's grade is satisfactory for progressing to the next level.
```

If the student's grade is less than or equal to 540 points, your program should print out something like

```
This student must improve his/her skills before progressing to the next level.
```

Note: use the 5 student_records listed above as test cases for your data.

Part 2: Working With Two-dimensional Lists

a. First write code to generate a two-dimensional list containing the first 100 numbers, starting at 1 and ending at 100. These numbers should be arranged in order, with 10 rows each containing 10 numbers. In other words, your list should look something like this:

```
number_list = [
      [1,2,3,4,5,6,7,8,9,10],
      [11,12,13,14,15,16,17,18,19,20].
…
      [91,92,93,94,95,96,97,98,99,100]
      ]
```

b. Now write code to sum each of the 10 rows. Print out the sum of the numbers in each row, clearly labeled. Your output should look something like this:

```
The sum of the numbers from 1 to 10 is: 55
The sum of the numbers from 11 to 20 is: 155
…
The sum of the numbers from 91 to 100 is: 955
```

c. Now write code to sum each of the 10 columns. That is, write code to sum 1+11+21+31+41+51+61+71+81+91. Print out that answer, clearly labeled, similar to:

```
The sum of column 0 is 460.
```

Then sum the other columns in the same way, printing out the answers:

```
The sum of column 1 is 470
The sum of column 2 is 480
…
The sum of column 9 is 550
```

Reminder

Assignments are your own effort. Do not share your code..

Grading Rubric

Part 1 works correctly: 40 points
Part 2 works correctly: 60 points

What to Submit

Your saved Python scripts. The easiest way to submit this assignment will be as two separate scripts. Please title your submission for Part 1 as `cw6_part1_(your_name).py` Please title your submissions for Part 2 as `cw6_part2_(your_name).py`.